Probit Regression Function

Applications in Government & Politics

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Topic 1 | Prerequisite Knowledge

Bernoulli Distribution

Used for modeling binary outcomes. applies
$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1-p)$

Logistic Regression

Probit is used instead of this model

Gradient

The vector of partial derivatives used to navigate the likelihood surface and find the maximum likelihood estimate.

Hessian

A matrix of second-order partial derivatives that describes the local curvature of the log-likelihood function.

Expected Hessian

Used in Fisher Scoring to simplify calculations; it is defined as being Positive Semi-Definite for the Probit model.

Link Function

Topic 2 | General Explanation

The Probit Mechanism

- **Distribution Assumption**: Assumes an underlying normal distribution for the latent variable.
- **Inputs**: Utilizes the CDF (Φ) and PDF (ϕ) of the standard normal distribution.
- **Predictor Matrix**: Uses a linear predictor $X\beta$, where X contains a column of 1s (intercept) and a column of observations.
- **Score Calculation**: Calculated using the difference between y variables and the normal PDF, scaled by the ratio of the PDF to the Bernoulli variance.
- **Iteration**: The system solves for β by using the Hessian and Gradient to update variables until convergence (difference $<$ tolerance).

Topic 3 | Mathematical Explanation

Topic 4 | Implementation

```r
myprobit = function(X, y, max_iter, tolerance, verbose=FALSE){

  m = ncol(X)
  beta <- rep(0, m)

  for (i in 1:max_iter) {

    eta <- X %*% beta

    phat <- pnorm(eta)
    qhat <- dnorm(eta)

    ratio <- qhat / (phat * (1 - phat))
    score <- (y - phat) * ratio
    g <- t(X) %*% score

    w <- (qhat^2) / (phat * (1 - phat))
    H <- -t(X) %*% (X * as.vector(w))

    copy_beta = beta
    beta <- beta - solve(H, g)

    if (verbose){
      cat("Iteration", i, ":\n")
      j=1
      for (b in beta){
        cat("Beta", j, " = ", beta[j], "\n")
        j=j+1
      }
    }
    if ((sum(beta - copy_beta)^2) < tolerance) {
        return(beta)
        break
      }
  }
}
```

```r
X <- model.matrix(am ~ mpg, data = mtcars)
y <- mtcars$am
myprobit(X,y,32, 0.00009, FALSE)
```

```
##                   [,1]
## (Intercept) -3.9100663
## mpg          0.1822605
```

Solving the System

Optimization Methods

Testing

Topic 5 | Comparison to Other Models

Motivation

Several methods exist for modeling binary outcomes. This section compares probit regression against three alternatives:

1. **Linear Regression (OLS)** — the standard baseline for continuous outcomes
2. **Logistic Regression** — the most widely used GLM for binary data
3. **Lasso Regression** — a penalized method that performs automatic variable selection

The goal is to understand the practical and theoretical differences between these approaches, and to identify when probit regression is the appropriate choice.

Data: Donner Party Survival

We use the **Donner Party survival dataset** from the `Sleuth3` package. The binary outcome is survival status (1 = Survived, 0 = Died), with Age and Sex as predictors.

```r
library(Sleuth3); library(dplyr); library(ggplot2)
library(glmnet);  library(caret)

set.seed(42)

# recode response and sex
donner <- case2001 %>%
  mutate(
    mystatus = if_else(Status == "Survived", 1, 0),
    male     = if_else(Sex == "Male", 1, 0)
  )

head(donner[, c("Age", "Sex", "Status", "mystatus")], 4)
```

```
##   Age    Sex   Status mystatus
```

Linear Regression on a Binary Response

OLS fits:

$$\hat{E}(Y \mid \mathbf{X}) = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots + \hat{\beta}_p X_p$$

When $Y \in \{0, 1\}$, this is the **Linear Probability Model (LPM)**. Since $\mathbf{X} \in \mathbb{R}$, fitted values are not constrained to $[0, 1]$, which means the model can produce predicted probabilities below 0 or above 1. This violates the basic requirement for a valid probability model and is the primary motivation for using a link function in GLMs.

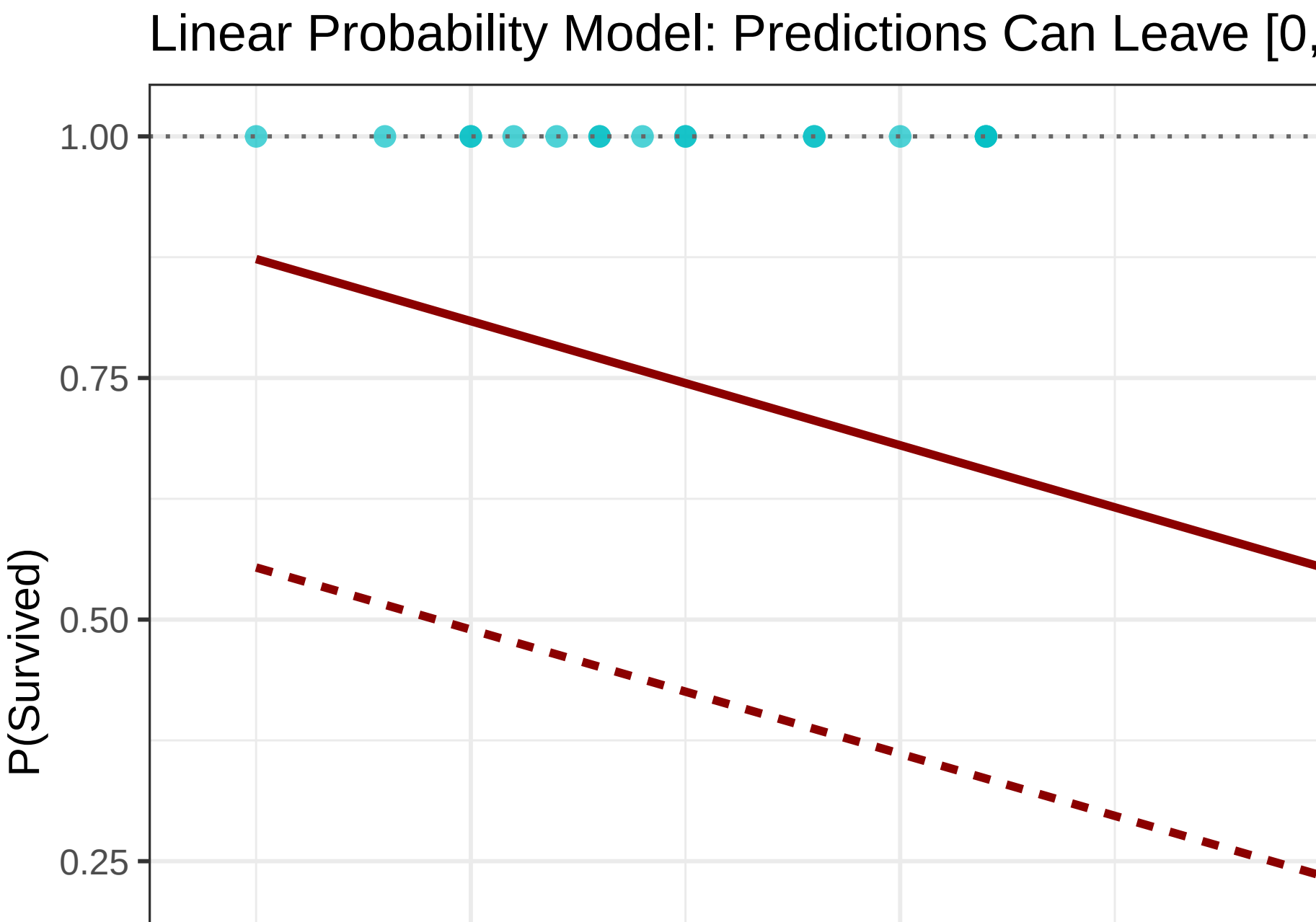
OLS on a Binary Response — Code

```r
# fit OLS
lpm_fit <- lm(mystatus ~ Age + male, data = donner)

donner <- donner %>%
  mutate(pi_lpm = predict(lpm_fit))

ggplot(donner, aes(x = Age, y = mystatus)) +
  geom_point(aes(color = Status), size = 2, alpha = 0.7) +
  geom_line(aes(y = pi_lpm, group = male,
                linetype = factor(male, labels = c("Female"
            color = "darkred", linewidth = 1) +
  geom_hline(yintercept = c(0, 1), linetype = "dotted", col
  labs(title = "Linear Probability Model: Predictions Can I
       y = "P(Survived)", x = "Age", linetype = "Sex",
       caption = "Dotted lines mark the valid [0,1] probabi
  theme_bw()
```

OLS on a Binary Response — Plot



OLS Predictions Outside [0, 1]

```r
lpm_preds <- predict(lpm_fit)

cat("Range of OLS predictions:", round(range(lpm_preds), 4)
```

```
## Range of OLS predictions: -0.0885 0.8731
```

```r
cat("Predictions below 0:     ", sum(lpm_preds < 0), "\n")
```

```
## Predictions below 0:      3
```

```r
cat("Predictions above 1:     ", sum(lpm_preds > 1), "\n")
```

```
## Predictions above 1:      0
```

```r
# misclassification at 0.5 threshold
lpm_class <- ifelse(lpm_preds > 0.5, 1, 0)
cat("OLS misclassification rate:",
    round(mean(lpm_class != donner$mystatus), 4), "\n")
```

Logistic Regression

Logistic regression uses the **logit link**:

$$\eta = \text{logit}(\pi) = \ln\left[\frac{\pi}{1-\pi}\right] = \mathbf{x}^T\boldsymbol{\beta} \quad \Rightarrow \quad \hat{\pi} = \frac{e^\eta}{1+e^\eta}$$

This assumes errors follow a logistic distribution.
Probit regression uses the **inverse standard normal CDF**:

$$\eta = \Phi^{-1}(\pi) = \mathbf{x}^T\boldsymbol{\beta} \quad \Rightarrow \quad \hat{\pi} = \Phi(\mathbf{x}^T\hat{\boldsymbol{\beta}})$$

This assumes errors follow a standard normal distribution. Both are valid GLMs for binary outcomes — they differ only in their distributional assumption.

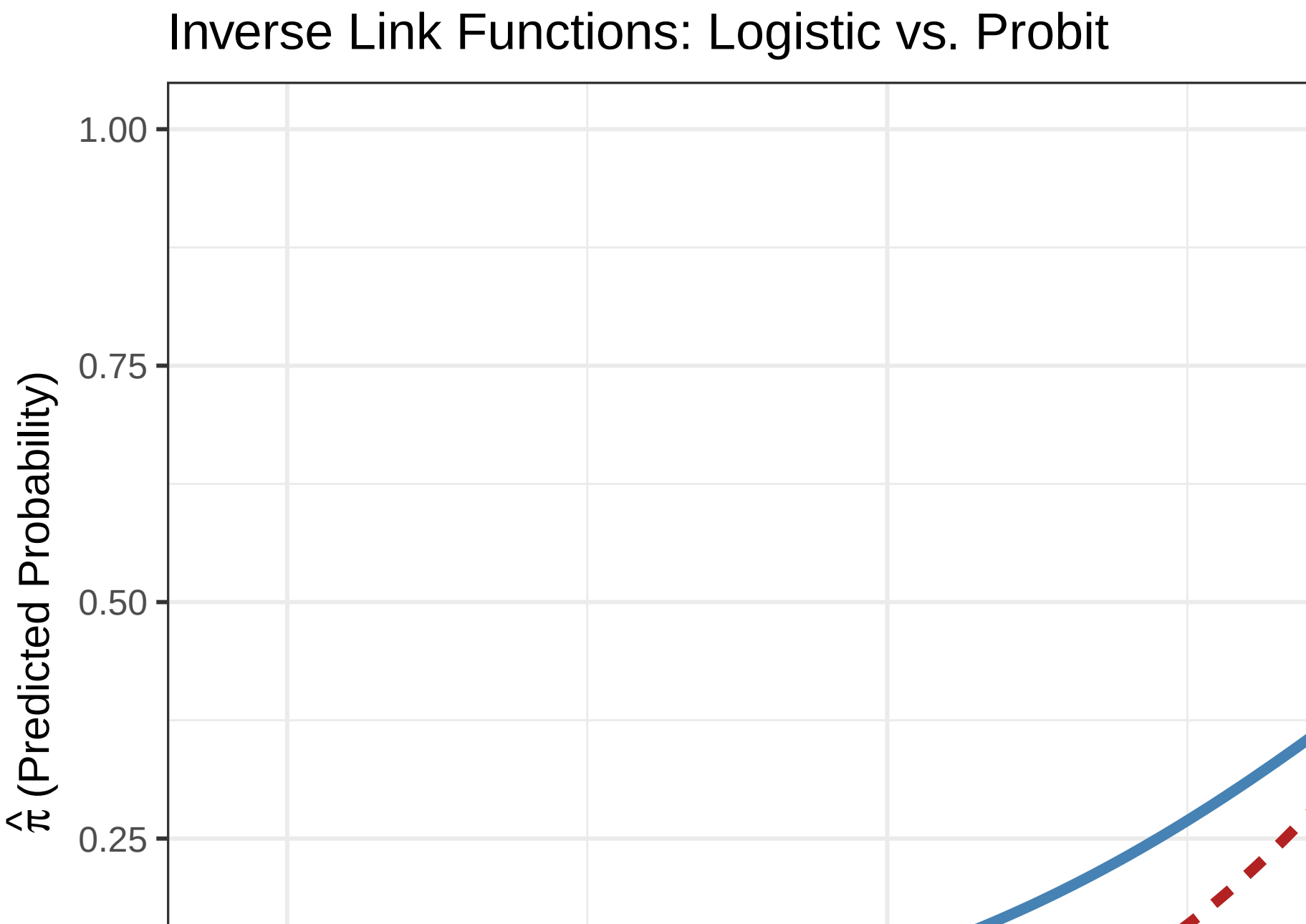
Link Functions — Code

```
eta_seq <- seq(-4, 4, length.out = 300)

df_links <- data.frame(
  eta    = rep(eta_seq, 2),
  pi     = c(plogis(eta_seq),  # inverse logit
             pnorm(eta_seq)),  # inverse probit (std norma
  Method = rep(c("Logistic (logit link)", "Probit (normal (
)

ggplot(df_links, aes(x = eta, y = pi, color = Method, linet
  geom_line(linewidth = 1.3) +
  labs(title = "Inverse Link Functions: Logistic vs. Probit
       x = expression(eta ~ "(Linear Predictor)"),
       y = expression(hat(pi) ~ "(Predicted Probability)"))
  scale_color_manual(values = c("steelblue", "firebrick"))
  theme_bw() + theme(legend.position = "bottom")
```

Link Functions — Plot



Fitting Both Models — Code

```r
# fit logistic and probit
logit_fit  <- glm(mystatus ~ Age + male, data = donner,
                  family = binomial(link = "logit"))

probit_fit <- glm(mystatus ~ Age + male, data = donner,
                  family = binomial(link = "probit"))

# compare coefficients
round(cbind(
  "OLS Coef"    = coef(lpm_fit),
  "Logit Coef"  = coef(logit_fit),
  "Probit Coef" = coef(probit_fit)
), 4)
```

Fitting Both Models — Output

```
##              OLS Coef Logit Coef Probit Coef
## (Intercept)   1.0658     3.2304      1.9173
## Age          -0.0128    -0.0782     -0.0457
## male         -0.3192    -1.5973     -0.9583
```

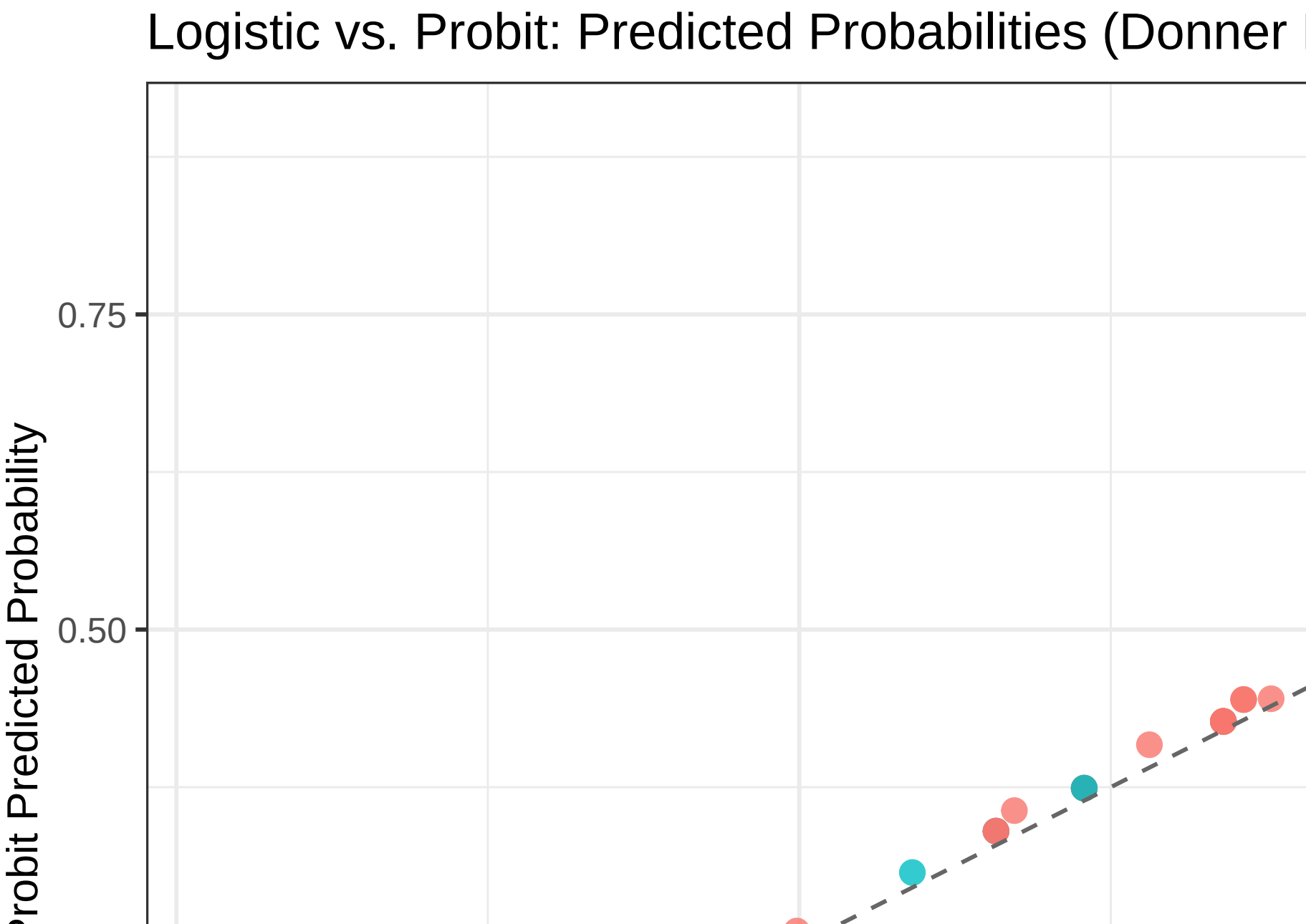
The probit coefficients are smaller by a factor of roughly 1.6 to 1.8. The logistic distribution has variance $\pi^2/3 \approx 3.29$, while the standard normal has variance 1. The scales differ, but the substantive conclusions are the same.

Predicted Probabilities — Code

```
donner <- donner %>%
  mutate(
    pi_logit  = predict(logit_fit,  type = "response"),
    pi_probit = predict(probit_fit, type = "response")
  )

ggplot(donner, aes(x = pi_logit, y = pi_probit, color = Sta
  geom_point(size = 2.5, alpha = 0.8) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed"
  labs(title = "Logistic vs. Probit: Predicted Probabilitie
       x = "Logistic Predicted Probability",
       y = "Probit Predicted Probability") +
  theme_bw()
```

Predicted Probabilities — Plot



Model Fit Metrics

```r
cat(sprintf("%-28s %8s %8s\n", "Metric", "Logistic", "Probi
```

```
## Metric                          Logistic   Probit
```

```r
cat(sprintf("%-28s %8.3f %8.3f\n", "AIC",
            AIC(logit_fit), AIC(probit_fit)))
```

```
## AIC                               57.256   57.283
```

```r
cat(sprintf("%-28s %8.3f %8.3f\n", "Residual Deviance",
            deviance(logit_fit), deviance(probit_fit)))
```

```
## Residual Deviance                 51.256   51.283
```

```r
cat(sprintf("%-28s %8.3f %8.3f\n", "Null Deviance",
            logit_fit$null.deviance, probit_fit$null.devian
```

```
## Null Deviance                     61.827   61.827
```

When to Prefer Probit

1. Latent variable framework. Assume an unobserved variable $Y^* = \mathbf{x}^T\beta + \varepsilon$ where $\varepsilon \sim N(0,1)$. We observe $Y = 1$ if $Y^* > 0$. This is the probit model exactly — a standard framework in economics and political science.

2. Tail behavior. The normal CDF approaches 0 and 1 more rapidly than the logistic distribution. Probit is preferable when probabilities near the extremes are of scientific interest.

3. Field convention. Probit is the standard choice in economics, political science, and related disciplines. For government and political science applications, probit is the conventional model.

4. Marginal effects. Probit coefficients do not have an odds-ratio interpretation, but average marginal effects provide a clear and directly interpretable quantity.

Marginal Effects

Probit coefficients have no direct probability interpretation on their own. The standard approach is to compute **Average Marginal Effects (AME)**:

$$\text{AME}_j = \frac{1}{n} \sum_{i=1}^{n} \phi(\mathbf{x}_i^T \hat{\beta}) \cdot \hat{\beta}_j$$

where $\phi(\cdot)$ is the standard normal PDF.

```r
library(margins)

# compute AMEs
ame <- margins(probit_fit)
summary(ame)[, c("factor", "AME", "SE", "z", "p")]
```

```
##  factor     AME     SE       z      p
##     Age -0.0148 0.0057 -2.6113 0.0090
##    male -0.3109 0.1188 -2.6160 0.0089
```

This gives the change in predicted probability for a one-unit change

Lasso Regression

The logistic or probit objective function can be **penalized** to perform variable selection:

$$\hat{\boldsymbol{\beta}} = \text{argmin}_{\boldsymbol{\beta}} \left\{ -\frac{2}{n}\ell(\boldsymbol{\beta}) + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

The ℓ_1 penalty shrinks small coefficients toward zero and eliminates irrelevant predictors entirely when λ is large enough. This is particularly useful in high-dimensional settings such as political science data with many demographic and behavioral predictors.

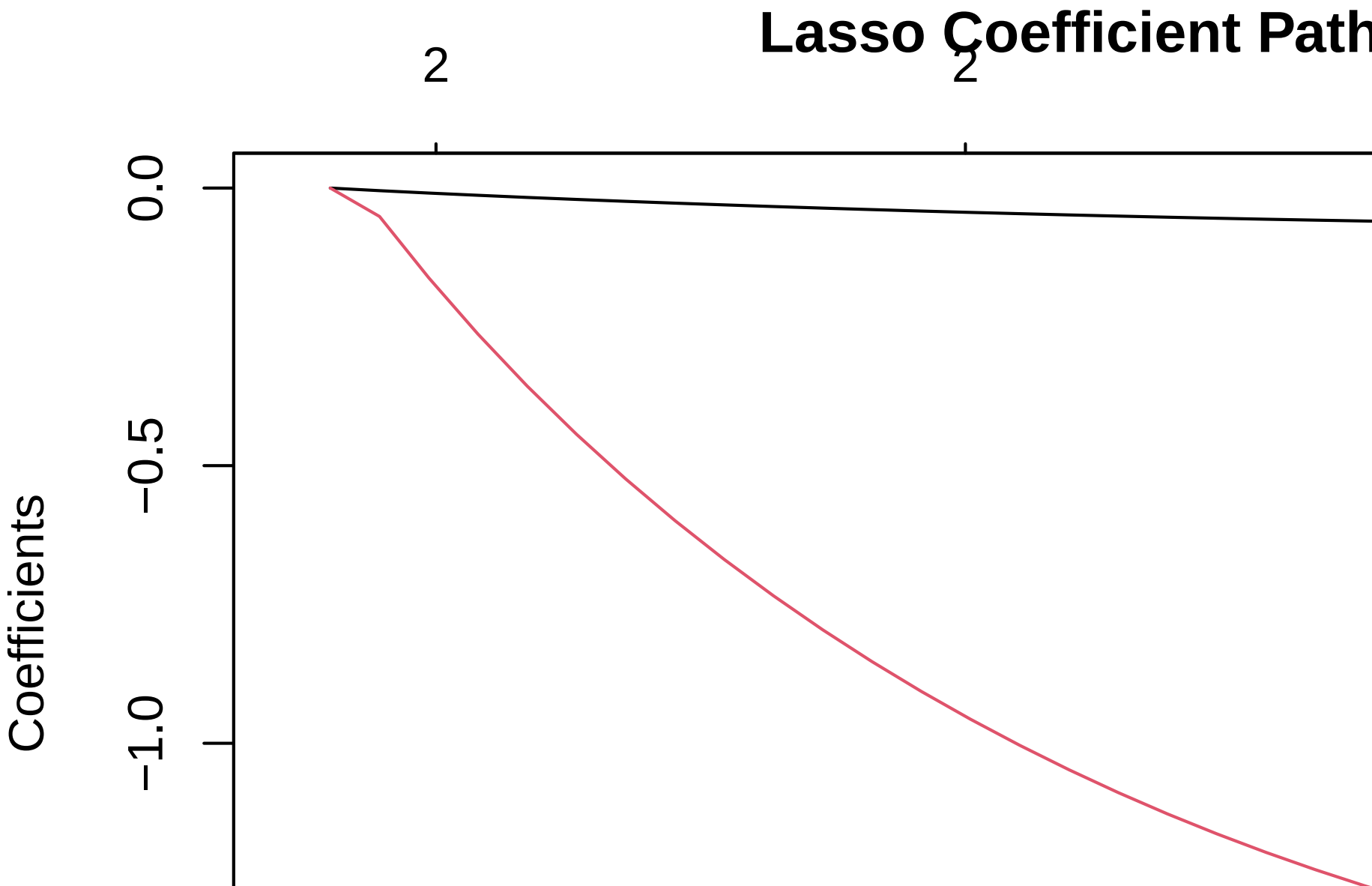
Lasso: Fitting the Model — Code

```
# glmnet needs a matrix, not a formula
X_mat <- model.matrix(mystatus ~ Age + male, data = donner)
y_vec <- donner$mystatus

# fit lasso over lambda path
lasso_path <- glmnet(X_mat, y_vec, family = "binomial", alp

plot(lasso_path, xvar = "lambda", label = TRUE,
     main = "Lasso Coefficient Path (Penalized Logistic, Do
```

Lasso: Coefficient Path



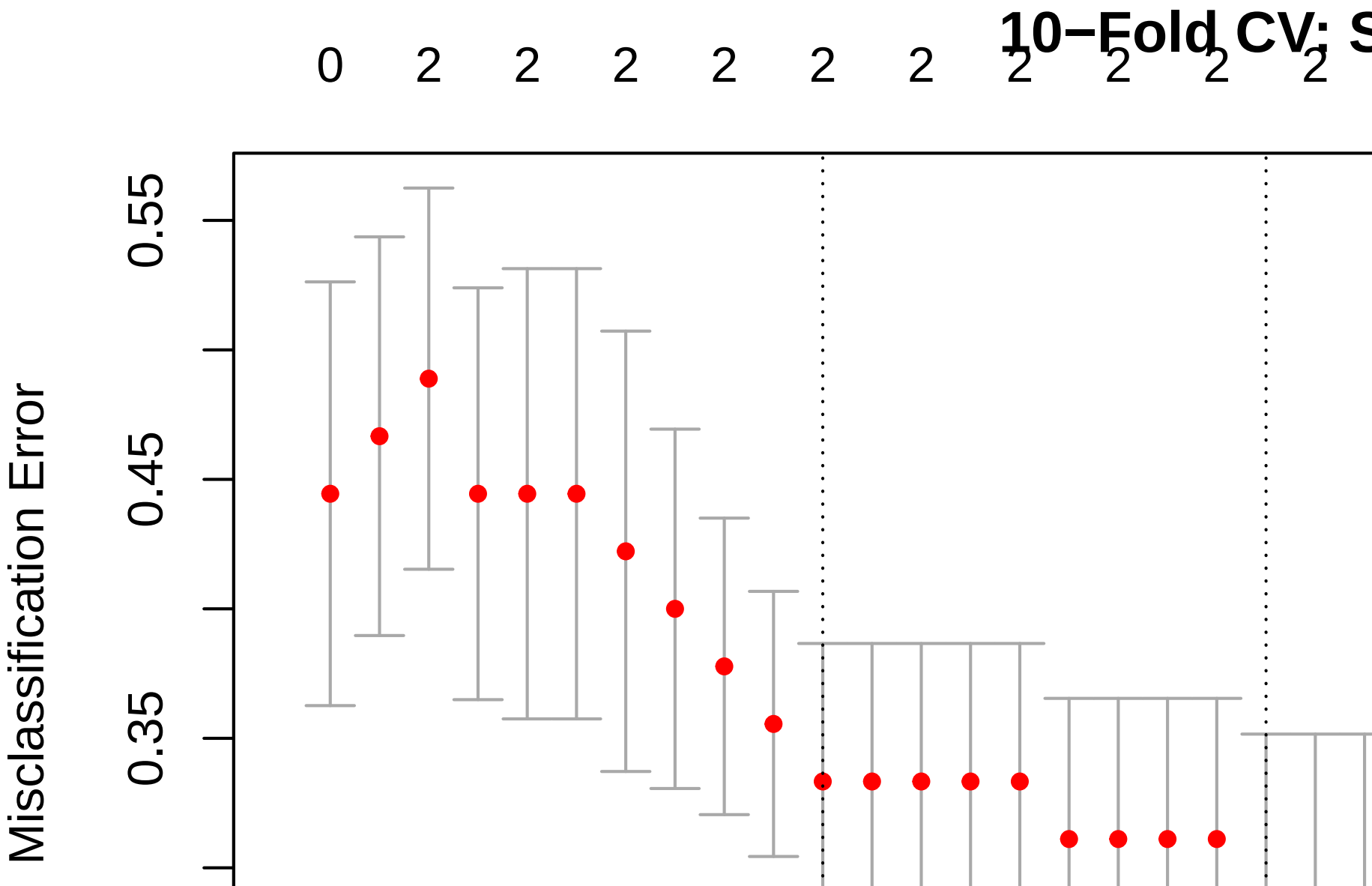
Lambda Selection via Cross-Validation — Code

```
set.seed(42)
cv_lasso <- cv.glmnet(X_mat, y_vec,
                      family       = "binomial",
                      alpha        = 1,
                      nfolds       = 10,
                      type.measure = "class")

plot(cv_lasso, main = "10-Fold CV: Selecting Optimal Lambda

cat("Lambda (min CV error): ", round(cv_lasso$lambda.min, 5
cat("Lambda (1 SE rule):    ", round(cv_lasso$lambda.1se, 5
```

Lambda Selection via Cross-Validation — Plot



Lasso Final Coefficients

```r
cat("Lasso coefficients at lambda.min:\n")
```

```
## Lasso coefficients at lambda.min:
```

```r
round(as.matrix(coef(cv_lasso, s = "lambda.min")), 5)
```

```
##              lambda.min
## (Intercept)    2.30441
## Age           -0.05615
## male          -1.19653
```

```r
cat("\nLasso coefficients at lambda.1se:\n")
```

```
##
## Lasso coefficients at lambda.1se:
```

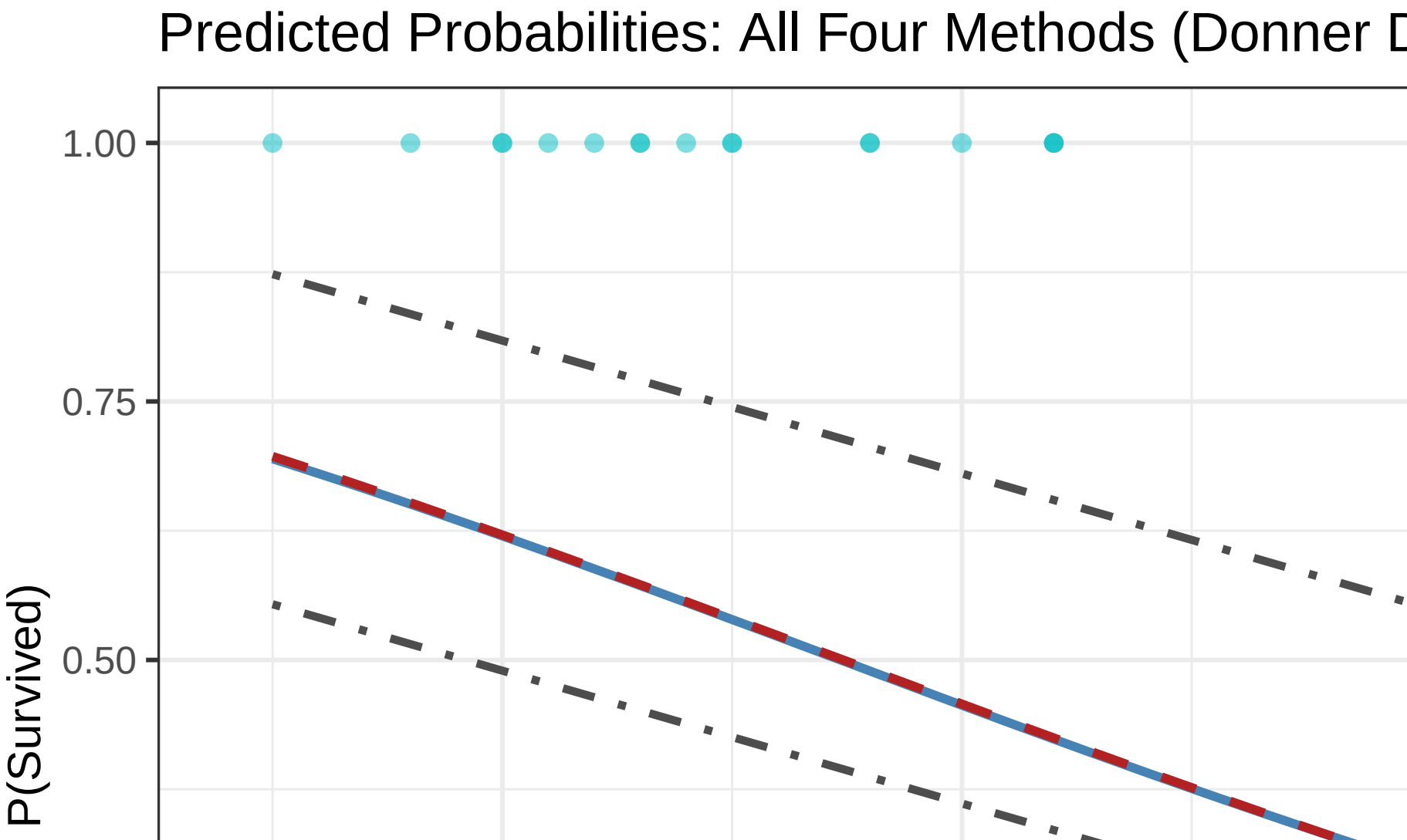
```r
round(as.matrix(coef(cv_lasso, s = "lambda.1se")), 5)
```

All Four Methods — Code

```
ggplot(donner, aes(x = Age)) +
  geom_point(aes(y = mystatus, color = Status), alpha = 0.5
  geom_line(aes(y = pi_lpm, group = male), color = "gray30"
            linewidth = 0.9, linetype = "dotdash") +
  geom_smooth(aes(y = mystatus), method = "glm",
              method.args = list(family = binomial(link = "
              se = FALSE, color = "steelblue", linewidth =
  geom_smooth(aes(y = mystatus), method = "glm",
              method.args = list(family = binomial(link = "
              se = FALSE, color = "firebrick", linewidth =
  labs(title = "Predicted Probabilities: All Four Methods (
       y = "P(Survived)", x = "Age",
       caption = "Blue = Logistic  |  Red = Probit  |  Gray
  theme_bw()
```

All Four Methods — Plot

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```



Confusion Matrix Comparison — Code

```
threshold <- 0.5

results <- donner %>%
  mutate(
    pred_lpm    = factor(ifelse(pi_lpm    > threshold, 1, (
    pred_logit  = factor(ifelse(pi_logit  > threshold, 1, (
    pred_probit = factor(ifelse(pi_probit > threshold, 1, (
    pred_lasso  = factor(ifelse(
      predict(cv_lasso, newx = X_mat, s = "lambda.min",
              type = "response")[, 1] > threshold, 1, 0)),
    actual = factor(mystatus)
  )

acc <- function(pred, actual) round(mean(pred == actual), 4

cat(sprintf("%-20s %s\n", "Method",    "Accuracy"))
cat(sprintf("%-20s %s\n", "OLS (LPM)", acc(results$pred_lpm
cat(sprintf("%-20s %s\n", "Logistic",  acc(results$pred_log
```

Confusion Matrix Comparison — Output

```
## Method                    Accuracy

## OLS (LPM)                 0.7556

## Logistic                  0.7778

## Probit                    0.7778

## Lasso                     0.7556
```

Probit Confusion Matrix

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  0  1
##          0 23  8
##          1  2 12
##
##                Accuracy : 0.7778
##                  95% CI : (0.6291, 0.888)
##     No Information Rate : 0.5556
##     P-Value [Acc > NIR] : 0.00166
##
##                   Kappa : 0.5361
##
##  Mcnemar's Test P-Value : 0.11385
##
##             Sensitivity : 0.6000
##             Specificity : 0.9200
```

Model Comparison Summary

Criterion	OLS (LPM)	Logistic	**Probit**	Lasso
Predictions in $[0, 1]$?	No	Yes	Yes	Yes
Link function	Identity	Logit	Φ^{-1} **(normal)**	Logit or Probit
Error distribution	Normal	Logistic	**Normal**	Logistic
Estimation method	OLS (closed form)	MLE (IRLS)	MLE (Fisher Scoring)	MLE + ℓ_1 penalty
Coefficient interpretation	$\Delta\hat{y}$	Log-odds / Odds ratio	Marginal effects (AME)	Log-odds (shrunk)
Variable selection?	No	No	No	**Yes**
Tuning required?	No	No	No	Yes (λ)

Key Takeaways

- **Linear regression (OLS)** is not appropriate for binary outcomes. Predicted values are not bounded to $[0, 1]$ and the model is misspecified.
- **Logistic and probit regression** produce nearly identical predicted probabilities and model fit. Coefficients differ in scale by approximately 1.6x due to differing error distributions, but substantive conclusions are the same.
- **Probit is preferred** in economics and political science due to the latent variable framework and field convention. Coefficients are interpreted via average marginal effects.
- **Logistic regression is preferred** when odds ratios are the natural effect measure.
- **Lasso** extends either model with an ℓ_1 penalty for automatic variable selection. Standard p-values and confidence intervals are not available after lasso selection.
- For government and political science applications, probit regression is the standard and appropriate choice.

References

Citations

Comparison Section (Rohan)

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- Abbott, M.G. ECON 452 Note 15: Marginal Effects in Probit Models. Queen's University. https://econ.queensu.ca/faculty/abbott/econ452/452note15.pdf
- Friedman, J., Hastie, T., Tibshirani, R. (2010). Regularization Paths for Generalized Linear Models via Coordinate Descent. *Journal of Statistical Software*, 33(1). https://glmnet.stanford.edu
- R Packages: `glmnet`, `caret`, `margins`, `Sleuth3`, `dplyr`,